DIPLOMA PROGRAMME PRACTICE · HL · SET A

Chemistry / Paper 2

Question booklet · 90 marks · 150 minutes

Answer every part. Show working, units and suitable precision. Use a calculator and a clean Chemistry data booklet. Use separate answer paper for additional working and diagrams.

QUESTION 1 [15]

Following a metal through two oxidation states

An aqueous first-row transition metal M has atomic number 26. A sample contains M^{2+} and M^{3+}. The two ions can be separated before their absorption spectra are measured.

(a) Write the noble-gas electron configurations of M, M^{2+} and M^{3+}. **[3]**

(b) Explain why the aquated ions may absorb different visible wavelengths even though the metal is the same. **[3]**

(c) A particular absorption has wavelength 510 nm. Determine the energy absorbed per mole of photons. Use $h=6.63\times10^{-34}$ J s, $c=3.00\times10^{8}$ m s^{-1} and $N_A=6.02\times10^{23}$ mol^{-1}. **[3]**

(d) The M^{3+} fraction of a 1.60 mmol mixture is first reduced completely to M^{2+} using 48.25 C, with 100% current efficiency. Use $Q=n(e^-)F$ and $F=96500$ C mol^{-1}. Determine the original amount of each ion. **[3]**

(e) A second analyst infers the original proportions from the mixed solution’s colour alone. Evaluate the inference and propose a quantitative alternative. **[3]**

QUESTION 2 [15]

Closing a magnesium oxide cycle

All values refer to 298 K in kJ mol^{-1}. Formation of MgO(s): −602; Mg atomization: +148; successive Mg ionizations: +738 and +1451; conversion of ½O_2(g) into O(g): +249; first and second O electron-gain enthalpies: −141 and +744.

(a) Write equations, including states, for the second ionization and each of the two electron-gain steps. **[3]**

(b) Determine the enthalpy of lattice dissociation, making the direction of the cycle explicit. **[3]**

(c) Explain why the second electron-gain step requires energy and why this does not prevent MgO from forming exothermically. **[3]**

(d) Predict how the magnitude of the lattice dissociation enthalpy changes on replacing Mg^{2+} with Ca^{2+}. Explain your reasoning. **[3]**

(e) An analyst uses +498 instead of +249 for the oxygen atomization step. Find the analyst’s dissociation enthalpy and explain the error. **[3]**

QUESTION 3 [15]

Substitution and stereochemical evidence

One enantiomer of 2-bromobutane reacts with water in a polar solvent. At constant solvent composition, doubling the substrate concentration doubles the initial rate. At constant substrate concentration, independently added nucleophile changes the initial rate negligibly. The substitution product has a small enantiomeric excess, defined here as the absolute difference between the percentages of the two enantiomers.

(a) State the empirical rate expression consistent with the data, give the units of k, and identify the bond broken in the slow step of the proposed unimolecular pathway. **[3]**

(b) Represent ionization and attack by water, describing the origin and destination of each electron-pair arrow. **[3]**

(c) Complete the mechanism and explain the formation of both configurations. **[3]**

(d) A sample contains 54.0% of one enantiomer and 46.0% of the other. Determine its enantiomeric excess and assess whether the data require a perfectly free carbocation. **[3]**

(e) Contrast this result with a single concerted backside displacement on the same enantiomer by OH^-. **[3]**

QUESTION 4 [15]

A shift reactor at 700 K

$CO(g)+H_2O(g) \rightleftharpoons CO_2(g)+H_2(g)$. Assume $\Delta H^\circ=-41.0$ kJ mol^{-1} and $\Delta S^\circ=-42.0$ J K^{-1} mol^{-1} are constant over the range considered. Use $R=8.31$ J K^{-1} mol^{-1}. Treat the gases as ideal.

(a) Calculate ΔG° at 700 K and deduce whether the standard-state forward reaction is product-favoured. **[3]**

(b) Calculate the equilibrium constant and state how changing the catalyst affects it. **[3]**

(c) Starting with 1.00 mol of each reactant and no products, calculate the equilibrium amount of H_2. **[3]**

(d) A different mixture at 700 K has $Q=15$. Calculate the actual ΔG and predict its initial direction of net change. **[3]**

(e) Find the temperature for $K=1$ and explain why compressing the equilibrium gas mixture at fixed temperature does not change its composition in this model. **[3]**

QUESTION 5 [15]

Iron content of a process sample

A 1.250 g process sample is dissolved without oxidizing its iron. All iron is present as Fe^{2+} in 250.0 cm^3. A 25.00 cm^3 aliquot consumes 20.00 cm^3 of 0.02000 mol dm^{-3} MnO_4^- in excess sulfuric acid. M(Fe)=55.85 g mol^{-1}. $E°(MnO_4^-/Mn^{2+})$=+1.51 V and $E°(Fe^{3+}/Fe^{2+})$=+0.77 V.

(a) Construct the balanced overall ionic equation using half-equations. **[3]**

(b) Calculate the mass percentage of iron in the original sample. **[3]**

(c) Determine the standard cell potential and $\Delta G°$ per mole of permanganate consumed. Use F=96500 C mol^{-1}. **[3]**

(d) A small fraction of Fe^{2+} oxidizes in air during dilution. Predict the changes to titre and calculated percentage, and suggest a control. **[3]**

(e) Each burette reading has uncertainty ±0.05 cm^3. Estimate the maximum percentage uncertainty from the titre and distinguish it from the oxidation error. **[3]**

QUESTION 6 [15]

Identifying a volatile carbonyl

A neutral compound with molecular formula $C_5H_{10}O$ shows a molecular ion at m/z 86, a strong IR absorption near 1715 cm^{-1} and no broad O–H absorption. Its ^1H NMR spectrum has only a triplet integrating to 6H and a quartet integrating to 4H. Assume ordinary first-order splitting.

(a) Calculate the index of hydrogen deficiency and explain which feature accounts for it. **[3]**

(b) Deduce the structure and assign both signals, including the reason for each splitting pattern. **[3]**

(c) Explain why pentan-2-one and pentanal each fail at least one observation and why the molecular-ion peak alone cannot distinguish them. **[3]**

(d) Predict the main product of reduction of the carbonyl by a suitable hydride reagent followed by protonation. Discuss chirality. **[3]**

(e) An unknown is proposed to be a mixture of unchanged ketone and its alcohol. Give two independent spectral observations expected for this mixture and explain why a broad O–H peak alone is insufficient. **[3]**